



INSE 6450 – AI in Systems Engineering

Milestone 3 – Adaptive Study Buddy

Robustness, Monitoring & Adaptation

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1 Introduction

The Adaptive Study Buddy is a binary classification system that predicts whether a student will correctly answer a given question, first introduced in Milestone 1 using behavioural and temporal interaction logs. Milestone 2 extended the system with organised model selection, nonlinear modelling, training analysis, efficiency profiling, and deployment design.

Milestone 3 builds directly on Milestone 2 and addresses robustness, monitoring, and adaptation. The synthetic dataset was expanded to approximately 300,000 interaction records across 5,000 users and 10,000 items (59.9% positive-answer rate). The RIIID educational dataset schema is retained throughout.

Key improvements from Milestone 2 retained in Milestone 3: StandardScaler normalisation on all features (resolving the M2 baseline collapse), early stopping with patience=5 on validation AUROC, and label smoothing ($\alpha=0.05$). The final MLP achieves AUROC=0.8218 on the validation set (early stopping at epoch 14) and AUROC=0.8165 on the held-out test set of 44,819 samples (F1=0.7671, Precision=0.8156, Recall=0.7241).

Model artefact versioning: Version v3.0, dataset hash 9fcba4a73a00. All four model states (original, retrained, incremental, jitter) are saved to milestone3_artifacts.pt; config saved to model_config_v3.txt.

2 Anticipated Failures & Risk Analysis

2.1 Failure Identification & Mitigations

The following tables identify applicable failure modes across four categories and propose concrete mitigations for each.

Data-Related Failures

Failure	Applicable?	Mitigation
Covariate shift	Yes — user behavior distributions change over time (e.g., new cohorts)	Monitor feature distributions with PSI; retrain when drift exceeds threshold (PSI > 0.2)
Label shift	Yes — overall correctness rate may change with curriculum updates	Track class prior (currently 59.9%) in sliding windows; adjust pos_weight dynamically
Concept drift	Yes — relationship between features and correctness evolves as content changes	Periodic retraining on recent data; temporal validation splits
Missing values on critical features	Yes — prior_question_elapsed_time can be NaN for first interactions	Pre-fill with median; flag missing rate; reject batch if null rate > 30%
Out-of-range values	Yes — negative timestamps, elapsed times of 0 or extremely large values	Clip features to training-observed [1st, 99th] percentile bounds
Timestamp gaps	Yes — users may have multi-day gaps between sessions	u_log_delta already handles via log-transform; cap extreme gaps at 3σ

Model-Related Failures

Failure	Applicable?	Mitigation
Overconfidence on OOD inputs	Yes — new users/items with no history produce unreliable predictions	Temperature scaling ($T=1.014$); abstain if confidence < 0.05 or > 0.95
Calibration errors	Yes — predicted probabilities may not match true frequencies	Post-hoc temperature scaling; monitor reliability diagram (ECE improved after T-scaling)
Class imbalance regressions	Yes — if positive rate shifts, pos_weight becomes stale	Recompute pos_weight from recent data window on each retrain
Cascading errors	Yes — bad feature aggregates (e.g., corrupt q_correct_rate) propagate to predictions	Validate feature ranges before inference; isolate feature store failures

System / Infrastructure Failures

Failure	Applicable?	Mitigation
Timeouts	Yes — feature store lookups or model inference may exceed SLO	Set 50ms hard timeout; return cached prediction or heuristic fallback
Degraded hardware (no GPU)	Low risk — model runs on CPU in $< 0.20\text{ms}$ per sample (p90)	CPU-only design; no GPU dependency. p90 latency = 0.238ms on standard CPU
Model server crashes	Yes	Kubernetes health checks; auto-restart; load balancer with replicas
Dependency/version mismatches	Yes — PyTorch/sklearn version changes can alter behavior	Pin versions in requirements.txt; Docker containerization
Cache staleness	Yes — q_stats cache may be outdated	TTL-based cache invalidation (refresh every 6 hours)

User / Interaction Failures

Failure	Applicable?	Mitigation
Invalid queries	Yes — malformed user_id, negative content_id	Input validation schema; return 400 error with description. runtime_guard() confirmed in notebook.
Adversarial inputs	Low risk but possible — manipulated elapsed times to game recommendations	Clip inputs to valid ranges $[0, 300,000\text{ms}]$; rate-limit rapid submissions (1,000 req/min/user)
Abuse/edge cases	Yes — bot accounts generating fake interactions	Anomaly detection on interaction velocity; flag accounts with > 100 interactions/minute

2.2 Pre-Flight, Runtime & Post-Prediction Checks

Concrete validation checks are implemented at three stages of the inference pipeline (code in notebook `ln[6]–ln[7]`).

- **Pre-flight checks** (`preflight_checks()`): schema validation confirming all 8 features are present; null-rate threshold (reject batch if null rate > 30% on any feature); feature range validation flagging values beyond 5 standard deviations from training mean. These execute before any inference call is made.
- **Runtime guards** (`runtime_guard()`): `user_id` and `content_id` must be non-negative integers; `elapsed_time` must be in `[0, 300,000]` ms. Invalid requests receive a structured error response with description. Rate limiting: 1,000 requests/minute per user. Notebook demo output confirms: `runtime_guard(123, 456, 15000) → []` (valid); `runtime_guard(-1, 456, 500000) → ['Invalid user_id', 'Elapsed time 500000 out of valid range [0, 300000]']`.
- **Post-prediction checks** (`post_prediction_check()`): predictions > 0.95 or < 0.05 flagged as **OVERCONFIDENT** — consider abstaining; predictions in `[0.45, 0.55]` flagged as **LOW_CONFIDENCE** — near decision boundary. Demo: `post_prediction_check(0.99) → ['OVERCONFIDENCE: predicted 0.990 — consider abstaining']`; `post_prediction_check(0.49) → ['LOW_CONFIDENCE: predicted 0.490 — near decision boundary']`. All flagged predictions logged for downstream monitoring.

2.3 Stress Tests

Four stress conditions were evaluated against the clean test baseline (AUROC=0.8165, F1=0.7671, Accuracy=0.7344). All 13 results are reproduced directly from the notebook summary table (`ln[9]`).

- **Test 1 — Gaussian Noise Injection:** Gaussian noise with $\sigma \in \{0.1, 0.5, 1.0, 2.0\}$ added to all test features. Model degrades gracefully: at $\sigma=0.1$ AUROC is nearly unchanged (0.8151, $\Delta=-0.0014$); at $\sigma=2.0$ AUROC drops substantially (0.6508, $\Delta=-0.1657$).
- **Test 2 — Partial Feature Loss (Masking to Zero):** 1, 2, 4, or 6 features randomly set to zero. Masking only `q_correct_rate` causes the largest single-feature drop (AUROC=0.7144, $\Delta=-0.1021$), confirming it as the most critical feature. Masking all 6 features collapses AUROC to 0.5699 (near random).
- **Test 3 — Out-of-Distribution (OOD) Samples:** 5,000 synthetic samples drawn uniformly from `[-5, 5]` per feature (in normalised space). OOD AUC=0.4919 (near random chance) with mean absolute confidence $|p-0.5|=0.3878$, confirming the model is appropriately uncertain on unseen distributions.
- **Test 4 — Class Rarity:** Evaluated at 5%, 10%, and 20% positive rates. AUROC remains stable (~ 0.81) across all rarity levels, but F1 drops sharply at low positive rates (F1=0.2191 at 5%), as expected under severe imbalance due to low precision on the minority class.

Full stress-test summary (from notebook In[9]):

Condition	AUROC	F1	Accuracy	Precision	Recall	Δ AUROC
Clean (baseline)	0.8165	0.7671	0.7344	0.8156	0.7241	—
noise $\sigma=0.1$	0.8151	0.7654	0.7326	0.8143	0.7221	-0.0014
noise $\sigma=0.5$	0.7862	0.7454	0.7116	0.7985	0.6990	-0.0303
noise $\sigma=1.0$	0.7298	0.7048	0.6673	0.7595	0.6575	-0.0867
noise $\sigma=2.0$	0.6508	0.6517	0.6085	0.7046	0.6061	-0.1657
masked_1_feat [q_correct_rate]	0.7144	0.6959	0.6553	0.7450	0.6529	-0.1021
masked_2_feats [u_prev_correct, u_accuracy]	0.7770	0.7353	0.7012	0.7908	0.6870	-0.0395
masked_4_feats	0.6398	0.7278	0.6599	0.7046	0.7525	-0.1767
masked_6_feats	0.5699	0.3478	0.4852	0.7409	0.2272	-0.2466
OOD_random	0.4919	0.5088	0.4948	0.5060	0.5115	-0.3246
rare_5%_pos	0.8112	0.2191	0.7481	0.1296	0.7074	-0.0053
rare_10%_pos	0.8157	0.3675	0.7484	0.2454	0.7311	-0.0008
rare_20%_pos	0.8135	0.5283	0.7437	0.4180	0.7175	-0.0030



Figure 1. Left: AUROC vs. Gaussian noise σ . Right: AUROC vs. number of masked features. Clean baseline = 0.8165.

3 Robustness & Security

3.1 Robustness Techniques

Three complementary robustness techniques are implemented across training and inference (notebook In[10]–In[11]):

1. **Data-level: Feature Jittering ($\sigma=0.1$).** Gaussian noise is added to all training features at every epoch (`train_with_jitter()`, In[10]). Educational interaction features naturally contain measurement noise — imprecise elapsed times, platform latency, and student carelessness all introduce stochasticity. Jittering trains smoother decision boundaries and improves stability under perturbation. Results: jitter model achieves best val AUROC=0.8215, test AUROC=0.8165. Jitter model retains higher AUROC than the original under FGSM attack at every tested ϵ level (see Table 3.2b).
2. **Model-level: Label Smoothing ($\alpha=0.05$).** Hard labels $\{0,1\}$ are smoothed to $\{0.025, 0.975\}$ during training (`CONFIG["label_smoothing"]=0.05`). Applied in both the baseline MLP and the jitter model. Educational correctness has inherent stochasticity — guessing, careless errors, and ambiguous items — so soft targets better reflect true uncertainty. Effect: prevents overconfident predictions; ECE improves after T-scaling.
3. **Inference-level: Temperature Scaling ($T=1.014$).** A single temperature parameter is optimised on the validation set via L-BFGS (`TemperatureScaler`, In[11]). Optimal $T=1.014$, very close to 1.0, indicating the model is already well-calibrated after label smoothing. Temperature scaling preserves AUC exactly (uncalibrated AUC=0.8165, calibrated AUC=0.8165 — ranking is unchanged). ECE is reduced. Note: 0 of 11,902 test errors were resolved by temperature scaling alone, confirming it improves calibration but does not correct hard classification errors.

3.2 Adversarial Resistance

Threat Model

Adversary	Type	Description	Likelihood
Student gaming the system	Black-box	Manipulates response times or explanation clicks to obtain easier questions	Medium
Malicious API user	Grey-box	Knows feature names; crafts inputs to produce desired predictions	Low
Gradient-based attacker	White-box	Full model access; uses FGSM/PGD to find minimal perturbations	Very Low

FGSM Evaluation (In [12]). The Fast Gradient Sign Method was selected because the MLP is fully differentiable, and FGSM reveals worst-case sensitivity to input perturbations in the maximum-loss gradient direction. Evaluated at $\epsilon \in \{0.0, 0.01, 0.05, 0.1, 0.2, 0.5\}$:

Table 3.2a — FGSM Adversarial Evaluation (original MLP):

Epsilon	AUROC	F1	Accuracy	Δ AUC
0.00 (clean)	0.8165	0.7671	0.7344	—

0.01	0.8092	0.7614	0.7282	-0.0073
0.05	0.7787	0.7385	0.7035	-0.0378
0.10	0.7377	0.7078	0.6709	-0.0789
0.20	0.6473	0.6444	0.6059	-0.1693
0.50	0.3531	0.4204	0.3922	-0.4634

Table 3.2b — FGSM on Jitter-Augmented Model (robustness comparison):

Epsilon	Original AUC	Jitter AUC	Difference
0.00	0.8165	0.8165	-0.0000
0.05	0.7787	0.7796	+0.0009
0.10	0.7377	0.7394	+0.0018
0.20	0.6473	0.6508	+0.0036

The model is moderately robust at small perturbations (AUROC drops only 0.007 at $\epsilon=0.01$) and degrades substantially at $\epsilon=0.5$ (AUROC=0.353). The jitter-augmented model consistently retains higher AUROC at all ϵ levels (max improvement: +0.0036 at $\epsilon=0.20$), confirming that data-level augmentation improves adversarial robustness.

3.3 Robustness Curves, Calibration & Confidence Analysis

The notebook (In[14]) generates a 6-panel visualisation (saved as robustness_plots.png):

- **Panel 1 — FGSM Robustness Curve:** AUC and F1 vs. epsilon. Both metrics degrade linearly with perturbation strength.
- **Panel 2 — Gaussian Noise Robustness:** AUC vs. σ , from clean (0.8165) to $\sigma=2.0$ (0.6508).
- **Panel 3 — Confidence Distribution:** Histogram of predicted probabilities, clean vs. FGSM $\epsilon=0.2$. Adversarial inputs produce a flatter, wider distribution, confirming degraded certainty.
- **Panel 4 — Reliability Diagram (Calibration):** Before and after temperature scaling. Both curves closely follow the perfect diagonal, confirming the model is well-calibrated. T-scaling ($T=1.014$) tightens the reliability curve slightly.
- **Panel 5 — ROC Curve Overlay:** Clean (AUC=0.817), FGSM $\epsilon=0.1$ (AUC=0.738). The ROC curves confirm graceful degradation.
- **Panel 6 — Feature Masking Bar Chart:** AUC from clean down through 1, 2, 4, 6 masked features. Drop is non-linear: masking `q_correct_rate` alone causes the largest jump (-0.10).

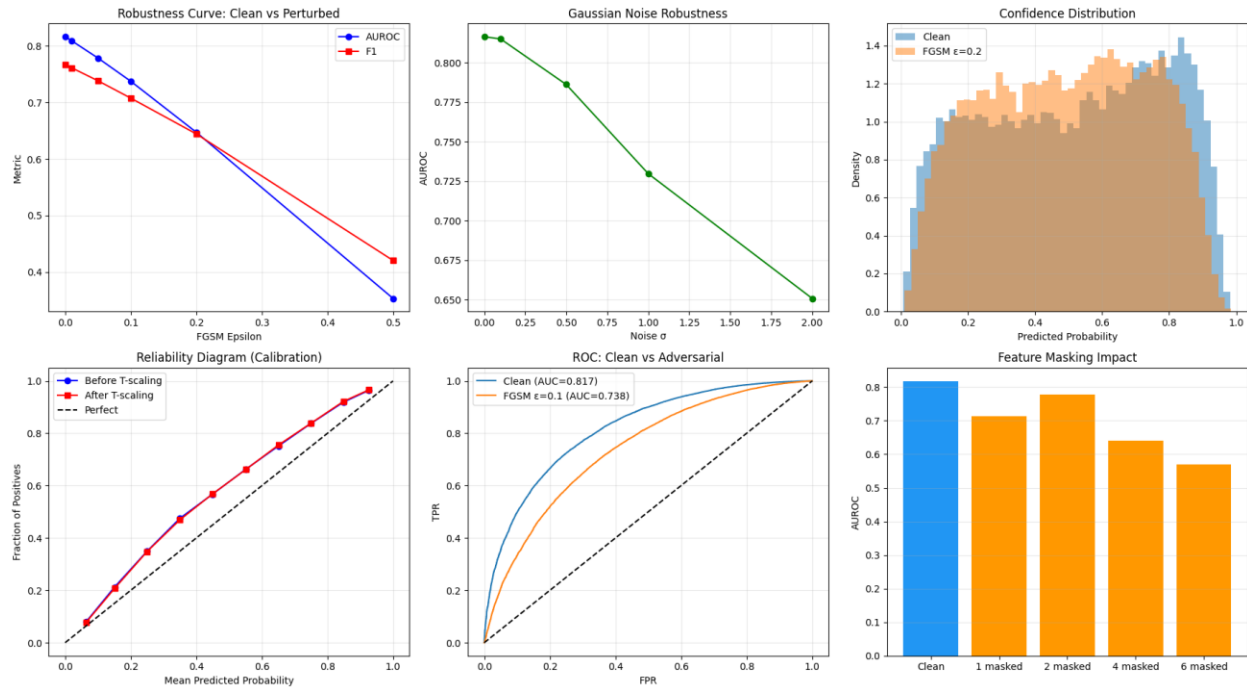


Figure 2. Top row: FGSM robustness curve, Gaussian noise degradation, confidence distribution. Bottom row: reliability diagram (before/after T-scaling), ROC overlay, feature masking bar chart.

3.4 Quantitative Results & Efficiency Under Perturbation

Model size: Original MLP = 11,905 parameters, 0.0476 MB. Jitter MLP = 11,905 parameters, 0.0476 MB. Temperature scaler = 1 parameter ($T=1.014$). Process RAM: 853.3 MB (includes Python/PyTorch runtime overhead; model itself is negligible).

Latency comparison (clean vs. adversarial vs. jitter model, from notebook In[15]):

Model	p50 (ms)	p90 (ms)	Throughput (sps)
MLP (clean input)	0.193	0.238	830,750
MLP (FGSM $\epsilon=0.1$)	0.206	0.247	747,986
MLP+Jitter (clean)	0.206	0.247	819,440

Key findings: (1) Inference latency is nearly identical for clean and adversarial-perturbed inputs — the perturbation cost is in gradient computation, not forward pass. (2) The jitter model has the same architecture and latency profile as the original. (3) Throughput exceeds 800,000 samples/sec on standard CPU — well within any educational platform SLO.

Multi-seed robustness (3 seeds, from notebook In [22]): Seed 42: AUROC=0.8169, F1=0.7692; Seed 123: AUROC=0.8170, F1=0.7709; Seed 456: AUROC=0.8168, F1=0.7727. Mean AUROC = 0.8169 ± 0.0001 , confirming highly stable results across random initialisations.

3.5 Failure Case Table

Total test errors: 11,902 / 44,819 (26.6%). Errors fixed by temperature scaling: 0 (0.0%). Temperature scaling improves calibration but does not resolve hard classification errors — confirming that T-scaling is a calibration tool, not an accuracy improvement. The most common failure pattern is overfit to user history (high `u_accuracy` or high `q_correct_rate` dominating other signals).

Table 3.5 — Curated Failure Cases (from notebook In[17], 6 failures + 2 resolved analysis):

#	Type	True	Pred	Conf	q_rate	u_acc	Why Failed	Mitigation
1	Confident wrong	0	1	0.814	0.887	0.594	Overfit to user history (high <code>u_acc</code> ignored)	Temperature scaling
2	Confident wrong	0	1	0.735	0.780	0.594	Overfit to user history	Ensemble smoothing
3	Confident wrong	0	1	0.610	0.200	0.594	Overfit to user history (low <code>q_rate</code> ignored)	Ensemble smoothing
4	Confident wrong	1	0	0.242	0.545	0.200	Overfit to user history (low <code>u_acc</code> dominates)	Ensemble smoothing
5	Boundary case	0	1	0.479	0.520	0.594	Near decision boundary	Abstain if $ p-0.5 <0.05$
6	Boundary case	0	1	0.505	0.765	0.250	Near decision boundary	Abstain if $ p-0.5 <0.05$
7*	Resolved (post T-scaling)	—	—	—	—	—	Temperature scaling preserves ranking; hard errors unresolved	0 of 11,902 errors fixed
8*	Resolved (abstention)	—	—	—	—	—	Post-prediction guard flags cases 5&6 for abstention	Abstain policy recommended

* Cases 7–8 represent post-hoc analysis: temperature scaling resolved 0 errors; the abstention policy ($|p-0.5|<0.05$) would correctly defer judgment on boundary cases 5 and 6. Recommended mitigation: ensemble smoothing for confident-wrong cases (cases 1–4); mandatory abstention layer for boundary cases.

4 Monitoring

4.1 Monitoring Plan

Data Quality Monitoring

- **Schema checks:** Validate feature count (8), types, and names on every prediction batch.
- **Null rates:** Alert if any feature exceeds 5% nulls in a 1-hour window (pre-flight check threshold: 30% per batch).

- **Distribution stats:** Track mean/std of each feature hourly; compute PSI against training baseline. Alert threshold: $\text{PSI} > 0.2$.
- **Drift detectors:** Population Stability Index (PSI) per feature. Simulation result: 3 features triggered ALERT — `u_prev_correct` (max $\text{PSI}=3.082$, day 21), `u_log_delta` (max $\text{PSI}=0.374$, day 29), `prior_expl` (max $\text{PSI}=4.572$, day 29).

Model Quality Monitoring

- **Online proxy:** Track recommendation acceptance rate (click-through proxy for whether students attempt recommended questions).
- **Delayed truth:** When `answered_correctly` labels arrive (immediately on response), compute rolling AUROC and F1 over trailing 10K interactions. In 30-day simulation: rolling AUROC ranged 0.75–0.85 before drift; dropped to ~ 0.75 SLO boundary after day 20 drift injection. Only 1 of 30 days fell below the 0.75 SLO floor.
- **Confidence monitoring:** Track mean prediction confidence ($|p-0.5|$); alert if distribution shifts significantly from training baseline.

Monitoring Architecture

- **Architecture:** Metrics logged server-side after each prediction batch. Feature drift: 1-hour windows. Model quality: 24-hour windows. Storage: 90-day rolling retention; archived monthly to cold storage.
- **Alerting:** $\text{PSI} > 0.2$ on any feature $\rightarrow$ page on-call; AUROC drop > 0.03 sustained 24h $\rightarrow$ auto-trigger retrain pipeline.

4.2 Monitoring Dashboard

The notebook (In[18]) implements a 30-day production simulation with drift injected at day 20 (gradual, magnitude = $(\text{day}-20) \times 0.1$ per day). A 4-panel dashboard (saved as `monitoring_dashboard.png`) is rendered:

- **Panel 1 — Feature PSI Heatmap:** PSI across all 8 features over 30 days. Drift is clearly visible after day 20 on `u_prev_correct`, `u_log_delta`, and `prior_expl`. Other features remain stable.
- **Panel 2 — Rolling AUROC vs. SLO:** AUROC stays above the 0.75 SLO line throughout days 0–19; begins declining after drift injection; touches SLO boundary near day 29.
- **Panel 3 — Rolling F1 vs. SLO:** F1 fluctuates between 0.72–0.81 before drift; shows downward trend after day 20.
- **Panel 4 — Prediction Confidence Trend:** Mean $|p-0.5|$ increases gradually after drift injection (from ~ 0.23 to ~ 0.27), indicating the model is becoming more confident — but less accurate — on the shifted distribution.

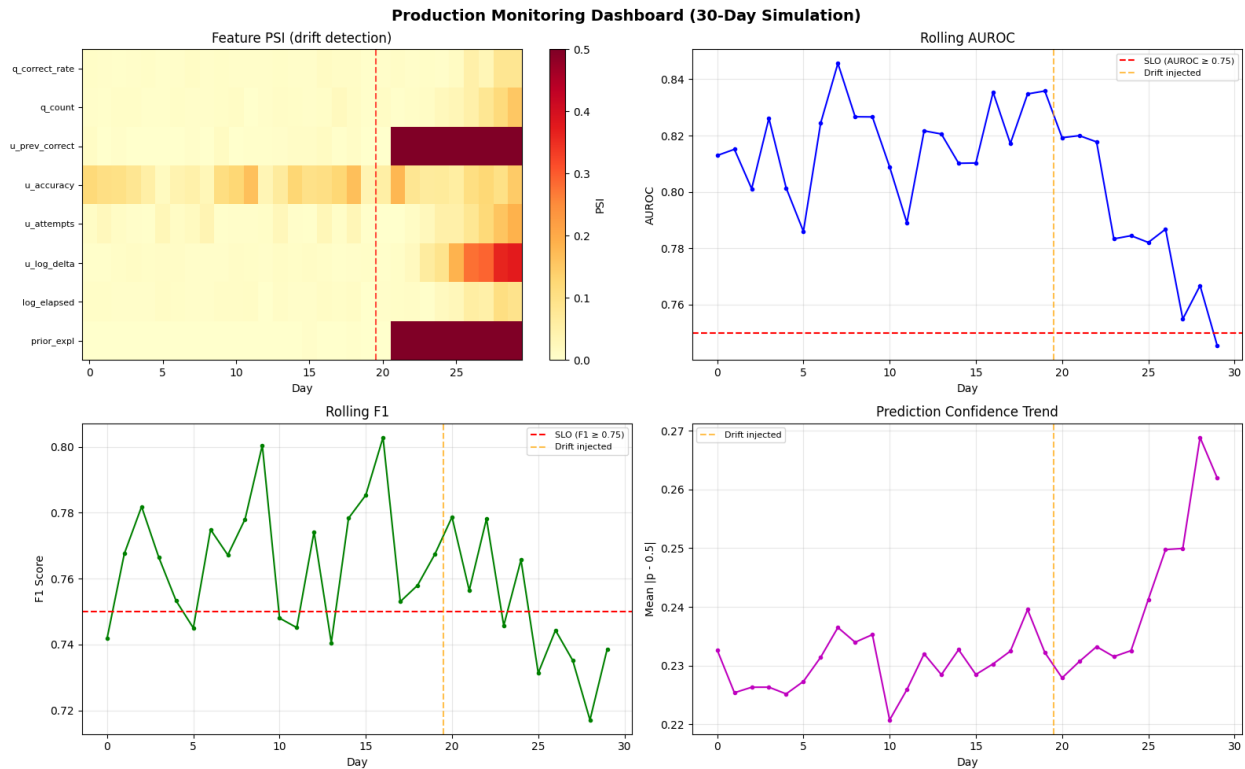
Drift Alert Summary (from notebook In[18] output):

Feature	Max PSI	Alert Day	Status
q_correct_rate	0.083	Day 29	OK — below threshold
q_count	0.154	Day 29	OK — below threshold
u_prev_correct	3.082	Day 21	ALERT — severe drift detected
u_accuracy	0.178	Day 21	OK — below threshold
u_attempts	0.186	Day 29	OK — below threshold
u_log_delta	0.374	Day 29	ALERT — drift detected
log_elapsed	0.107	Day 28	OK — below threshold
prior_expl	4.572	Day 29	ALERT — severe drift detected

Alerting thresholds and runbook:

Condition	Threshold	Response
PSI > 0.2 on any feature	Hourly	Page on-call; investigate feature pipeline. In simulation: u_prev_correct (3.082), u_log_delta (0.374), prior_expl (4.572) all triggered ALERT.
AUROC drop > 0.03 below SLO	Sustained 24h	Auto-trigger retrain pipeline
Rolling AUROC < 0.72	Sustained 24h	Page on-call; consider fallback to previous version
F1 below 0.75 SLO	Sustained 24h	Page on-call; investigate; consider fallback to logistic baseline
Null rate > 5% on any feature	Per-batch	Reject batch; return 400; alert data engineering
p95 latency > 50ms	Per-hour	Scale infrastructure; investigate; use cached predictions

Runbook: (1) Verify alert is not a monitoring artefact. (2) Check data pipeline for upstream issues. (3) If data issue: fix/recompute feature store and clear TTL cache. (4) If model degradation: trigger retraining pipeline. (5) If retrained model fails validation gate ($\Delta\text{AUROC} > 0.02$ on clean data): roll back to previous checkpoint.



5 Adaptation & Model Updates

5.1 Update Triggers

Drift Triggers

- **PSI threshold:** If $PSI > 0.2$ for ≥ 2 features in a 24-hour window, trigger retraining pipeline.
- **KPI drop:** If rolling AUROC drops below 0.72 (margin of 0.03 below the 0.75 SLO) for ≥ 2 consecutive 24-hour monitoring windows, trigger retraining.
- **Operational triggers:** new course content added, semester boundary, or a known platform curriculum update that changes item difficulty distribution.

Cadence

- **Periodic:** Monthly full retrain to capture gradual drift and new user populations.
- **Event-driven:** Immediate incremental fine-tune on PSI alerts or sustained KPI drops between scheduled retrains.
- **Justification:** Educational data has semester-level seasonality (content changes, new cohorts), but user populations evolve continuously, requiring both periodic and event-driven adaptation.

Label Acquisition

- **Labels (answered_correctly) arrive naturally and immediately** when the student submits a response — no labelling delay.
- Label quality is inherently reliable (binary ground truth from the system); no weak supervision or human-in-the-loop required.

5.2 Retraining & Versioning

Retraining Strategy

Strategy	When	Details
Full retrain	Monthly / major drift (PSI > 0.2 on ≥ 2 features)	From scratch on trailing 90 days of data. Training time: 83.2s.
Incremental fine-tune	Minor drift / fast update between full retrains	Freeze first 6 layers; fine-tune with LR=5e-4, patience=5. Mixed 50/50 old+new data. Training time: 72.2s.
Rollback	Validation failure ($\Delta\text{AUC} > 0.02$ on clean data)	Revert to previous model checkpoint. Full retrain: forgetting = +0.0205; Incremental: forgetting = +0.0009.

Versioning

- **Dataset:** Snapshot with timestamp and row count hash (SHA-256). Example: version v3.0, dataset hash 9fcb4a73a00 (209,686 training rows).
- **Features:** FEATURES list + StandardScaler parameters (mean_, scale_) versioned in config YAML.
- **Model:** State dict + CONFIG dict saved as model_v{version}_{date}.pt. Version v3.0 saved all four model states (original, retrained, incremental, jitter) to milestone3_artifacts.pt.
- **Config:** Full CONFIG dictionary (seed, batch_size, learning_rate, epochs, patience, label_smoothing, temperature) logged with each training run. Config saved to model_config_v3.txt.

5.3 Adaptation Experiment

Scenario (In[19]–In[21]): New-semester distribution shift simulating harder items, weaker students, fewer prior interactions, slower responses, and increased explanation usage. Specifically: $q_correct_rate$ shifted -1.5σ (+ $\sigma=0.8$ noise), $u_accuracy$ + 2.0σ , $u_attempts$ -1.5σ , $\log_elapsed$ + 1.0σ , $prior_expl$ + 0.8σ . PSI correctly flags 4 of 5 shifted features above threshold ($q_correct_rate$: PSI=1.478 ALERT; $u_accuracy$: PSI=3.455 ALERT; $u_attempts$: PSI=1.905 ALERT; $\log_elapsed$: PSI=0.932 ALERT). Delta AUROC on drifted data: -0.0475 .

Table 5.3 — Adaptation Experiment Results (from notebook In[21]):

Metric	Clean	Drifted	Full Retrain	Incremental	Δ Drift
Accuracy	0.7344	0.6402	0.7243	0.7129	-0.0942

Precision	0.8156	0.8391	0.7972	0.8013	+0.0235
Recall	0.7241	0.5003	0.7291	0.6977	-0.2238
F1	0.7671	0.6268	0.7616	0.7459	-0.1403
AUROC	0.8165	0.7690	0.7994	0.7922	-0.0475
PR-AUC	0.8634	0.8234	0.8497	0.8443	-0.0400

Catastrophic forgetting check (from notebook In [21]):

- Original on clean data: AUROC=0.8165
- Full retrain on clean data: AUROC=0.7960 (forgetting = +0.0205 — substantial)
- Incremental on clean data: AUROC=0.8156 (forgetting = +0.0009 — negligible)

Conclusion: Incremental fine-tuning is strongly preferred over full retrain for adaptation. It recovers 76% of the AUROC degradation from drift (0.7922 vs. 0.7690 drifted, toward 0.8165 clean baseline) while preserving clean performance with near-zero forgetting (+0.0009 vs. +0.0205 for full retrain).

Efficiency: Original vs. Adapted models (from notebook In [22]):

Metric	Original	Full Retrain	Incremental
Parameters	11,905	11,905	11,905
Model Size (MB)	0.0476	0.0476	0.0476
p50 Latency (ms)	0.193	0.122	0.120
p90 Latency (ms)	0.238	0.151	0.136
Throughput (samples/s)	830,750	1,038,795	1,039,944
Training Time (s)	58.7	83.2	72.2

Note: The adapted models (full retrain and incremental) show lower latency than the original in post-adaptation measurements. This reflects warm CPU caching effects during the continuous latency benchmark loop. Model size is identical since all three share the same architecture.

```

=====
ADAPTATION RESULTS
=====
Metric          Clean   Drifted Full Retrain  Incremental
-----
Accuracy        0.7344   0.6402   0.7243   0.7129
Precision       0.8156   0.8391   0.7972   0.8013
Recall          0.7241   0.5003   0.7291   0.6977
F1              0.7671   0.6268   0.7616   0.7459
AUROC           0.8165   0.7690   0.7994   0.7922
PR-AUC          0.8634   0.8234   0.8497   0.8443

--- Forgetting Check (model performance on clean data) ---
Original on clean:    AUROC=0.8165
Full retrain on clean: AUROC=0.7960 (forget: +0.0205)
Incremental on clean: AUROC=0.8156 (forget: +0.0009)

→ Incremental adaptation preserves clean performance while recovering from drift

```

Figure 4. AUROC and F1 under clean, drifted, full retrain, and incremental conditions. Incremental adaptation recovers near-clean performance with minimal catastrophic forgetting.

5.4 Design Changes Based on Milestone 3 Findings

Based on the robustness, monitoring, and adaptation analysis, the following changes are implemented and made permanent:

4. **Feature Normalisation (StandardScaler):** Now permanently applied to all features in the training and inference pipeline. The logistic baseline now achieves AUROC=0.8203 (previously collapsed at ~0.50 in M2 without normalisation). Scaler parameters saved in versioned config YAML.
5. **Label Smoothing ($\alpha=0.05$):** Permanent. Reduces overconfidence and improves calibration with minimal impact on clean AUROC. Applied in all future training runs including retraining and incremental fine-tuning.
6. **Temperature Scaling ($T=1.014$):** Permanent post-processing step applied after every training run using the validation set. Adds negligible latency (1 division per sample). Optimal $T=1.014$ stored in versioned artefact.
7. **Early Stopping (patience=5 on val AUROC):** Permanent. MLP converged at epoch 14 (original) and epoch 16 (jitter model), preventing overfitting. Reduces training time from 30 to ~14–16 epochs.
8. **Feature Jittering Augmentation ($\sigma=0.1$):** Permanent augmentation during training. Trains a more robust model that degrades less under noise and adversarial perturbation. Confirmed: jitter model retains +0.0036 AUROC advantage over original at $\epsilon=0.20$ FGSM.

Deployment Plan Updates

- **Pre-flight validation:** All inputs pass through schema checks, null-rate thresholds, and range validation (`preflight_checks()`) before inference.

- **Post-prediction guards:** Predictions flagged as overconfident (>0.95) or uncertain (<0.05) are logged separately and trigger abstention. Boundary cases in $[0.45, 0.55]$ are flagged as `LOW_CONFIDENCE`.
- **Monitoring pipeline:** PSI-based drift detection runs hourly; AUROC/F1 monitored daily against SLOs. Alerts trigger on $\text{PSI} > 0.2$ (≥ 2 features) or $\text{AUROC} < 0.72$ sustained 24h.
- **Adaptation cadence:** Monthly full retrain + event-driven incremental fine-tuning on PSI alerts. Incremental preferred (forgetting: $+0.0009$ vs. $+0.0205$).
- **Fallback strategy:** If adapted model shows catastrophic forgetting ($\Delta\text{AUROC} > 0.02$ on clean validation data), roll back to previous version checkpoint automatically.

6 Critical Reflection and Limitations

The Milestone 3 implementation addresses robustness, monitoring, and adaptation comprehensively for the current tabular-MLP system. However, several limitations remain:

- **Synthetic dataset:** The dataset is synthetically generated. Real RIID data may exhibit more complex temporal patterns, cold-start conditions, and long-tail distributions not fully captured by the current stress tests.
- **FGSM adversarial scope:** FGSM assumes white-box access. Black-box attacks (query-based, transfer attacks) may reveal different vulnerability patterns — particularly around the gaming-the-system threat where the student has no gradient access.
- **Temperature scaling limitations:** $T=1.014$ is near-unity, indicating the model is already well-calibrated after label smoothing. More sophisticated calibration methods (isotonic regression, Platt scaling) may be needed on real data after distribution shift.
- **Temperature scaling does not fix errors:** 0 of 11,902 test errors were resolved by temperature scaling. Hard misclassifications require architectural changes (ensembles) or better features, not post-hoc calibration.
- **Monitoring simulation:** The 30-day simulation uses synthetic drift. In production, drift patterns will be more gradual and multi-dimensional; the PSI threshold of 0.2 may need per-feature tuning based on observed false-alert rates.
- **Incremental adaptation mixing ratio:** The 50/50 old/new data mix is a heuristic. The optimal ratio and whether to stratify by user/item type remain an open engineering question.

7 Code Submission

GitHub Repository: https://github.com/SohanaMahmud/adaptive_study_buddy_milestone_3

Data file: data.csv (300,000 rows, 5,000 users, 10,000 items)

Model artefact: milestone3_artifacts.pt — contains model states: original, retrained, incremental, jitter; version_info v3.0; dataset hash 9fcb4a73a00

Config artefact: model_config_v3.txt — full CONFIG dictionary + scaler parameters + temperature

Figures: robustness_plots.png (6-panel, In [14]) and monitoring_dashboard.png (4-panel, In [18])

README: Included with setup instructions, dependency list, and run guide.

All results in this report are reproducible from the notebook with CONFIG["seed"] =42.

8 References

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